

Prospectus

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In 1980 Vera Rubin published a paper that is credited with being the first substantial evidence for dark matter (Rubin, Ford, and Thonnard 1980). However, no one has been able to identify the unknown substance beyond speculation. Reasonable efforts have failed to describe the rotational curves Rubin first noticed, which implies a new direction needs to be tried.

When a photon travels through a plasma medium its energy is conserved. Which means that the plasma frequency and wave number of the photon must change inversely, where the plasma frequency is dependant on the density of the plasma. This behavior means that as a photon travels through a varying density plasma, if the plasma density goes up the wave number of the photon must go down. Using Snell's law the photon's angle

Bibliography

Rubin, V. C., W. K. Ford Jr., and N. Thonnard (June 1, 1980). “Rotational Properties of 21 SC Galaxies with a Large Range of Luminosities and Radii, from NGC 4605 (R=4kpc) to UGC 2885 (R=122kpc).” In: *The Astrophysical Journal* 238, pp. 471–487. ISSN: 0004-637X. DOI: 10.1086/158003. URL: <https://ui.adsabs.harvard.edu/abs/1980ApJ...238..471R> (visited on 04/15/2026).